

Reviewer Pack — Minimal Local Ring Norm in Algebraic Geometry and Communicat...

Entry f259ada99140 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-01 16:55:27 UTC

Minimal Local Ring Norm in Algebraic Geometry and Communication Complexity Rank

Entry ID: f259ada99140 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-01 16:55:27 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry over finite fields **Field B** (complexity object): Communication Complexity

Statement:

For a Boolean function f , the minimal norm of a local ring associated with its algebraic variety is $O(f(n^{1/2}))$ in terms of the communication complexity rank r_f .

Rationale (proposer's reasoning):

The minimal norm of a local ring can be computed using techniques from algebraic geometry, which are relatively underexplored in complexity theory. This invariant captures the geometric structure of the algebraic variety and may reveal hidden relationships with communication complexity, a well-established field within complexity theory.

Taxonomy category: Algebraic_Geometry (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): fba8f12ace13042a

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the ratio of the minimal norm of a local ring to $n^{1/2}$ is less than or equal to $O(f(n^{1/2}))$ for all 30 random seeds, with a correlation coefficient greater than 0.7.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	SAFE	HITS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - title:(Minimal Local Ring Norm Algebraic Geometry finite fields) AND abstract:communication complexity - subject:Algebraic Geometry over finite fields AND subject:Communication Complexity AND rank:algebraic variety norm - algorithm:local ring norm AND algorithm:communication complexity rank AND $f(n^{1/2})$

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=242.2s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity_rank(f):
        n = len(f)
        max_consecutive_ones = 0
        current_count = 0
        for bit in f:
            if bit == 1:
                current_count += 1
                max_consecutive_ones = max(max_consecutive_ones, current_count)
            else:
                current_count = 0
        return max_consecutive_ones

    def minimal_local_ring_norm(f):
        n = len(f)
        # Simplified approximation for demonstration purposes
        return Fraction(n**2, 4)

    n_values = [5, 10, 15, 20, 30, 40]
    total_metric_value = 0
    instances_tested = 0
    max_n = 0

    for n in n_values:
```

```

f = generate_boolean_function(n)
r_f = communication_complexity_rank(f)
norm = minimal_local_ring_norm(f)

if norm <= 0 or r_f <= 0:
    continue

ratio = norm / (n**Fraction(1, 2))
total_metric_value += ratio
instances_tested += 1
max_n = max(max_n, n)

if instances_tested < 30:
    return {
        "metric_name": "Ratio of Norm to sqrt(n)",
        "metric_value": None,
        "instances_tested": instances_tested,
        "n_max": max_n,
        "conjecture_holds": False,
        "counterexample": "Too few instances tested"
    }

mean_metric_value = total_metric_value / instances_tested
return {
    "metric_name": "Ratio of Norm to sqrt(n)",
    "metric_value": mean_metric_value,
    "instances_tested": instances_tested,
    "n_max": max_n,
    "conjecture_holds": False,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all("conjecture_holds" in r and r["conjecture_holds"] for r in results):
        support_fraction = sum(1 for r in results if "conjecture_holds" in r and r["conjecture_holds"])
        RESULT = f"SUPPORTED mean={sum(r['metric_value'] for r in results) / len(results)} std=0.0"
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not ("conjecture_holds" in result and result["conjecture_holds"]))
        RESULT = f"FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}"

    print(RESULT)

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means that the pre-registered support conditions could not be evaluated. | next: Re-run the test with increased time limits to ensure it completes and produces results for further analysis.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	39031
2	preregistration	ollama_remote	glm4:latest	0	0	14600
3	novelty	ollama_remote	glm4:latest	0	0	11586
4	novelty	ollama_remote	glm4:latest	0	0	46478
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11440
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	33429
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10933
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22233
9	judge	ollama_remote	glm4:latest	0	0	24431

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 214162 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/f259ada99140.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/f259ada99140.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/f259ada99140.tar.gz* (if generated)